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SUPERSEDING
NAVORD OS 5845D
15 October 1957

MILITARY SPECIFICATION

POWER SUPPLY MARK 20 MOD 1

1. SCOPE

1.1 This specification establishes the requirements for the procurement of the Power Supply Mark 20 Mod 1. The power supply is used with the Test Set Mark 26 Mods 0, 1, and 2 and Test Set Mark 96 Mods 0 and 1.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	Boxes, Fiber.

MILITARY

MIL-P-116	Preservation, Methods of.
MIL-B-131	Barrier Material; Water Vapor-proof, Flexible.
MIL-T-152	Treatment, Moisture-and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment.
MIL-V-173	Varnish, Moisture and Fungus Resistant for the Treatment of Communications, Electronic and Associated Electrical Equipment.

FSC 6625

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MIL-D-3464	Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification.
MIL-S-6872	Soldering Processes, General Specification for.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-122	Color Code for Chassis Wiring for Electronic Equipment.
MIL-STD-129	Marking for Shipment and Storage.

DRAWINGS

BUREAU OF ORDNANCE

LD 166645	Power Supply Mark 20 Mod 1.
LD 284866	Preparation for Delivery for Test Set Cable Assemblies.
1476194	Marking Instructions for Storage and Shipping Containers.

And all drawings listed in the above LD's and drawing.

PUBLICATIONS

BUREAU OF ORDNANCE

NAVORD OP 400	General Instructions for Design, Manufacture and Inspection of Naval Ordnance Equipment.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specifice procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements.

3.1.1 The Power Supply Mark 20 Mod 1 shall be manufactured in accordance with LD 166645 and all documents listed thereon.

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3.1.2 Preproduction sample. - Unless otherwise specified in the contract or order, a preproduction sample shall be manufactured by the same procedures and processes and at the same location as proposed for full production. This sample will be tested for the purpose of determining compliance with the requirements of this specification, the drawings, and the contract. See 4.1.2.

3.1.3 Interchangeability. - Component parts of assembled units (unless otherwise specified on the applicable drawings) fabricated in strict agreement with their applicable drawings shall be interchangeable and shall assemble with mating components of the unit without selection or fitting. If, during assembly of the fabricated parts of a unit, selection or fitting is required and changes in dimensions are necessary, no changes in the parts as delineated on applicable drawings shall be made without prior approval of the Bureau of Ordnance.

3.2 Moisture-and fungus-resistant treatment. - Materials and parts of this power supply which are not moisture and fungus resistant shall be treated, before assembly, with varnish conforming to Military Specification MIL-V-173. The application of the moisture-and fungus-resistant varnish shall be in accordance with MIL-T-152.

3.3 Requests for deviation. - Requests for deviation from this specification, applicable drawings, specifications, publications, materials and processes specified herein, shall be as set forth in OP 400.

3.4 Conflicting requirements. - Conflicting requirements arising between this specification and any other specifications, publications, or drawings listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation and clarification.

3.5 Soldering. - Unless otherwise specified, soldering processes shall conform to the requirements specified in Military Specification MIL-S-6872. Only non-corrosive and non-volatile fluxes shall be used. Rosin shall be used when soldering wires to contacts and terminals. The use of rosin core solder is permitted. Electrical connections shall be mechanically secure and electrically continuous before soldering. Sufficient heat shall be applied during the soldering process to avoid "cold" solder joints.

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3.6 Wiring. - All internal wiring shall be color coded in conformance with MIL-STD-122 and shall be harnessed or laced where possible.

3.7 Performance requirements and product characteristics. - The power supply covered by this specification shall meet the following performance requirements and product characteristics.

3.7.1 Insulation resistance. - The insulation resistance between electrical circuits and between each circuit and the chassis shall be not less than 50 megohms when tested as specified in 4.5.3.

3.7.2 Meter adjustment. - With the METER SWITCH in each of the positions shown below, the following meter indications shall be obtainable, by adjustment, as specified in 4.5.4.

METER SWITCH POSITION	METER INDICATION
OFF	0
202	40
141	40
-67.5	40
141A	40
141B	40
141C	40
105	38-42
-48	27 Min.

3.7.3 Voltage output. - The power supply shall give an output as shown below, within the following tolerances, when tested as specified in the indicated paragraphs.

PARA.	VOLTAGE	VOLTAGE TOLERANCE AT NO LOAD	DEVIATION OF LOADED VOLTAGE FROM NO-LOAD VOLTAGE
4.5.5	202	±10	±1
4.5.6	141	±7	±1.5
4.5.7	141A	±7	±3
4.5.8	141B	±7	±3
4.5.9	141C	±7 volts	±3 volts
4.5.10	-67.5	±3.5	±1
4.5.11	105	±5	±1
4.5.12	-48	38 Min.	±10
4.5.13	40 (AC)	±5	±5

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3.8 Workmanship and cleanliness. - The power supply shall be free from dust, dirt, metal filings, moisture, solder splashes, chips, and other foreign matter. Necessary precautions shall be taken during manufacture and assembly to insure these conditions.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Government inspection. - The Government inspector will make such inspections as are necessary to determine that all components and assemblies are in accordance with the requirements of the contract, pertinent specifications and drawings. Unless otherwise specified, these inspections will be conducted in accordance with instructions contained in OP 400.

4.1.1 Sampling. - Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government inspector in the determination of the acceptability of products submitted by a supplier for Government inspection will be in accordance with the provisions of MIL-STD-105.

4.1.2 Preproduction sample. - Unless otherwise specified for each contract or order, a representative sample of one power supply shall be manufactured using the methods and procedures proposed for full production. The Government inspector shall perform, witness, or participate in such inspections and tests on the sample as are required for full production. Upon conditional acceptance by the inspector, the preproduction sample shall be forwarded, together with a report of inspection, to an activity designated in the contract or by the procuring agency, for the purpose of determining compliance with the requirements of the contract, specifications and drawings. Further production of the item by the contractor prior to approval of the contracting activity or the completion of the tests on preproduction samples shall be at the contractor's risk. Preproduction samples accepted will be applied as part of the quantity specified by the contract or order.

4.1.2.1 Preproduction sample for a subsecuent contract. - The necessity for a preproduction sample will be determined by the procuring agency when production under a new contract by the same contractor at the same location follows the manufacture of satisfactory material as covered by this document.

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4.1.3 Lot. - As applied to the Government inspection of units of product, the term "lot" shall mean "inspection lot", i.e. a collection of units of the product used as a basis for Government inspection. Unless otherwise specified, the number of units of product in "inspection lots" shall be as determined by the Government inspector, and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

4.2 Contractor's inspection. - Contractor's inspection shall be conducted in accordance with provisions of "Inspection by Contractor" as contained in OP 400.

4.3 Classification of tests. - The inspection and testing of mechanisms covered by this document shall be classified as follows:

(a) acceptance tests

(b) preproduction tests

4.3.1 Acceptance tests. - Acceptance tests for acceptance of the power supply covered by this specification shall consist of tests to determine compliance with all requirements specified in this document.

4.3.2 Preproduction tests. - Preproduction tests are those performed on samples submitted in accordance with 4.1.2 to determine that the lot and production meet all the requirements of this specification. Acceptance shall be based on no defects in a sample.

4.4 Test equipment.

4.4.1 Test and inspection equipment and facilities. - Unless otherwise specified, the manufacturer shall furnish and maintain all necessary test equipment, facilities and personnel for performing all acceptance tests under the supervision of the Government inspector. The test equipment shall be adequate in quantity, and unless definite requirements are specified, they shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

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4.4.1.1 Equipment required for performing these tests is as follows:

(a) 115-volt, 60-cycle a-c supply

(b) d-c voltmeter, 0-300 volts, accuracy within ±1/2 percent, 5000 ohms per volt sensitivity.

(c) a-c voltmeter, 0-150 volts, accuracy within ±1/2 percent, 1000 ohms per volt sensitivity.

(d) One 4-megohm 1/2 watt resistor, accuracy within ±10 percent.

(e) One 10,000-ohm, 5 watt resistor, accuracy within ± 10 percent.

(f) One 1,000-ohm, 10-watt resistor, accuracy within ±10 percent..

(g) One 2,000-ohm, 10-watt resistor, accuracy within ±10 percent.

(h) Megohmmeter-500 volts d-c. Range 10.0-100 megohms, accuracy within ±10 percent.

4.5 Tests.

4.5.1 Test conditions. - Unless otherwise specified, tests shall be performed at 80° ±10°F and ambient relative humidity.

4.5.2 Visual inspection. - The power supply and its component parts shall be inspected to verify that the materials, design, construction, physical dimensions, soldering and workmanship are in accordance with this document and the drawings.

4.5.3 Insulation resistance. - The power supply shall be subjected to a conditioning period of 24 hours at a temperature of 77° ±2°F, and a relative humidity of 75 ±5 percent. A test potential of 500 ±50 volts d-c is then applied, using test item 4.4 1.1 (h), as specified in 3.7.1. The insulation resistance shall be not less than 50 megohms.

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4.5.4 Meter adjustment test.

(a) Connect plug on power cable to 115-volt, 60-cycle a-c supply, test item 4.4.1.1 (a), and set ON-OFF switch at ON. Set primary voltage selector switch at 115V.

(b) Set METER SWITCH at OFF. Meter shall indicate zero.

(c) Set METER SWITCH at 202 V. By turning 202 V. ADJ. control, it shall be possible to cause meter to indicate 40.

(d) Set METER SWITCH at 141 V. By turning 141 V. ADJ. control, it shall be possible to cause meter to indicate 40.

(e) Set METER SWITCH at -67.5V. By turning -67.5V. ADJ. control, it shall be possible to cause meter to indicate 40.

(f) Set meter switch at 141 AV, 141 BV and 141 CV; in each case meter shall indicate 40. Set meter switch at 105V; meter shall read between 38 and 42. Set meter switch at -48V; meter shall read at least 27.

(g) Set METER SWITCH at OFF.

4.5.5 202-volt output test.

(a) Set METER SWITCH at 202 V.

(b) Turn 202 V. ADJ. control until test set meter indicates 40.

(c) Connect negative terminal of external voltmeter, test item 4.4.1.1 (b), to jack J2 below ON-OFF switch and connect positive terminal to contact 2 of 12-contact socket on rear of power supply. Voltmeter shall indicate 202 volts ±10 volts.

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(d) Remove negative terminal from jack and connect to contact 8 of 12-contact socket. Voltmenter shall indicate 202 volts ±10 volts.

(e) Remove positive terminal from contact 2 and connect to jack J3 below 202 V. ADJ. control. Voltmeter shall indicate 202 volts ±10 volts.

(f) Connect 4-megohm resistor, test item 4.4.1.1 (d), across voltmeter terminals. Voltmeter shall indicate voltage of step (e) ±1 volt.

(g) Remove voltmeter and 4-megohm resistor.

4.5.6 141-volt output test.

(a) Set METER SWITCH at 141 V.

(b) Turn 141 V. ADJ. control until test set meter indicates 40.

(c) Connect negative terminal of external voltmeter, test item 4.4.1.1 (b), to contact 8 of 12-contact socket and connect positive terminal to jack J4 below 141 V. ADJ. control. Voltmeter shall indicate 141 volts ±7 volts.

(d) Remove positive terminal from jack and connect to contact 12 of 12-contact socket. Voltmeter shall indicate 141 volts ± 7 volts.

(e) Connect 2,000-ohm, 10-watt resistor, test item 4.4.1.1 (g), across voltmeter terminals. Voltmeter shall indicate voltage of step (d) ±1.5.

(f) Remove voltmeter and resistor.

4.5.7 141-A-volt output test.

(a) Set METER SWITCH at 141-A V. Meter shall indicate 40.

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(b) Connect negative terminal of voltmeter, test item 4.4.1.1 (b), to contact 8 of 12-contact socket and connect positive terminal to contact 3. Voltmeter shall indicate 141 volts ±7 volts.

(c) Connect 10,000-ohm, 5-watt resistor, test item 4.4.1.1 (e), across voltmeter terminals. Voltmeter shall indicate voltage of step (b) ±3 volts.

(d) Remove voltmeter and 10,000-ohm resistor.

4.5.8 141-B-volt output test.

(a) Set METER SWITCH at 141-B V. Meter shall indicate 40.

(b) Connect negative terminal of voltmeter; test item 4.4.1.1 (b), to contact 8 of 12-contact socket and connect positive terminal to contact 5. Voltmeter shall indicate 141 volts ±7 volts.

(c) Connect 10,000-ohm resistor, test item 4.4.1.1 (e), across voltmeter terminals. Voltmeter shall indicate voltage of step (b) ±3 volts.

(d) Remove voltmeter and resistor.

4.5.9 141-C-volt output test.

(a) Set METER SWITCH at 141-C V. Meter shall indicat 40.

(b) Connect negative terminal of voltmeter, test item 4.4.1.1 (b), to contact 8 of 12-contact socket and connect positive terminal to contact 6. Voltmeter shall indicate 141 volts ±7 volts.

(c) Connect 10,000-ohm resistor, test item 4.4.1.1 (e), across voltmeter terminals. Voltmeter shall indicate voltage of step (b) ±3 volts.

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(d) Remove voltmeter and 10,000-ohm resistor.

4.5.10 -67.5-volt output test.

(a) Set METER SWITCH at -67.5 V.

(b) Turn -67.5 V. ADJ. control until meter indicates 40.

(c) Connect positive terminal of voltmeter, test item 4.4.1.1 (b), to contact 8 of 12-contact socket and connect negative terminal to jack below -67.5 V. ADJ. control. Voltmeter shall indicate 67.5 volts ±3.5 volts.

(d) Remove negative terminal from jack and connect to contact 9 of 12-contact plug. Voltmeter shall indicate 67.5 volts ±3.5 volts.

(e) Connect 10,000-oh, 5-watt resistor, test item 4.4.1.1 (e), across voltmeter terminals. Voltmeter shall indicate voltage of step (d) ±1 volt.

(f) Remove voltmeter and resistor.

4.5.11 105-volt output test.

(a) Set METER SWITCH at 105 V. Meter shall indicate between 38 and 42.

(b) Connect positive terminal of voltmeter, test item 4.4.1.1 (b), to contact 1 of 12-contact socket and connect negative terminal to contact 4. Voltmeter shall indicate 105 volts ±5 volts.

(c) Connect 10,000-ohm, 5-watt resistor, test item 4.4.1.1 (e), across voltmeter terminals. Voltmeter shall indicate voltage of step (b) ±1 volt.

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(d) Remove voltmeter and resistor.

(e) Connect positive terminal of voltmeter to the chassis and the negative terminal to contact 1 of the 12-contact socket. Voltmeter shall indicate less than 0.5 volt.

(f) Connect positive terminal of voltmeter to the chassis and the negative terminal to contact 4 of the 12-contact socket. Voltmeter shall indicate less than 0.5 volt.

(g) Remove voltmeter.

4.5.12 -48-volt output test.

(a) Set METER SWITCH at -48 V. Meter shall indicate at least 27.

(b) Connect positive terminal of voltmeter, test item 4.4.1.1 (b), to contact 8 of 12-contact plug and connect negative terminal to contact 11. Voltmeter shall indicate at least 48 volts.

(c) Connect 1,000-ohm, 10-watt resistor, test item 4.4.1.1 (f), across voltmeter terminals. Voltmeter shall indicate at least 38 volts.

(d) Remove voltmeter and resistor.

4.5.13 A-c output test.

(a) Set METER SWITCH at MAIN RECTIFIER.

(b) Connect a-c voltmeter, test item 4.4.1.1 (c), across contacts 7 and 10 of 12-contact socket. Voltmeter shall indicate 40 volts ±5 volts.

(c) Connect 1,000-ohm, 10-watt resistor, test item 4.4.1.1 (f), across voltmeter. Voltmeter shall indicate voltage of step (b) ±5 volts.

(d) Remove voltmeter and resistor.

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4.6 Acceptance criteria. - Failure of any power supply to meet any of the requirements and tests of this specification shall result in rejection of the power supply. If the number of failing power supplies exceeds the number permitted by the associated criteria, the lot represented will be rejected.

5. PREPARATION FOR DELIVERY

NOTE: This equipment is very fragile. Contractor shall contact Bureau of Ordnance (Mab-1g) for special handling instructions).

5.1 Preservation and packaging.

5.1.1 Level A.

5.1.1.1 Cleaning, drying, and preservative application. - (Not applicable).

5.1.1.2 Unit packaging. - Prior to unit packaging, the power supply cables shall be packaged in accordance with Bureau of Ordnance List of Drawings LD 284866 and positioned in cable well of power supply. Each power supply shall be initially packaged in a Style RSC fiberboard box fabricated from Class 1, Type I or Type II board, having 275 lb. minimum bursting strength, conforming to the requirements of Federal Specification PPP-B-363. Sufficient fillers and folded corner pads, of the same box material, shall be utilized within the carton to provide a tight pack and to prevent direct contact between protrusions of the power supply case and the walls of the fiberboard box.

5.1.1.2.1 Each power supply, prepared as specified above, shall be further packaged Method IIb conforming to the requirements of Military Specification MIL-P-116. The inner carton, containing the packaged power supply, shall be inserted into a water-vaporproof bag, fabricated from Class I barrier-material conforming to requirements of Military Specification MIL-B-131, and closure heat-sealed.

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5.1.1.3.2 A Style RSC fiberboard box, Class 2, W5c or W5s compliance symbol, conforming to the requirements of Federal Specification PPP-B-636, shall be employed for the outer carton and designed to snugly enclose the barrier-sealed inner container. The required desiccant shall be suitabley positioned within the power supply case. For conformance and quantity required see Military Specification MIL-D-3464.

5.1.2 Level B. - (Not applicable)

5.1.3 Level C.

5.1.3.1 Power supply units shall be packaged in a manner which will afford adequate protection from deterioration and physical damage during shipment from the contractor or supply source to the first receiving activity for immediate use.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior containers. - Each power supply, individually packaged as specified under 5.1.1, shall be packed in an overseas type wood-cleated plywood box or a Class 2, Style 2, 2 1/2, 3 or 4 nailed wood box conforming to the requirements of Federal Specification PPP-B-601 or PPP-B-621, respectively.

5.2.2 Level B. - (Not applicable)

5.2.3 Level C.

5.2.3.1 Power supply units shall be packed in a manner which will provide protection from damage during direct shipment from the contractor or supply source to the first receiving activity for immediate use. The type of containers used shall conform to applicable carrier rules and regulations.

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5.3 Marking.

5.3.1 Normal markings. - In addition to any special marking required by the contract or order, all marking shall be in accordance with Military Standard MIL-STD-129.

5.3.1.1 For details on marking requirements, see Bureau of Ordnance Drawing Number 1476194.

5.3.1.2 Special markings. - The following special marking shall appear on the top, one (1) side and one (1) end of each shipping container:

DELICATE INSTRUMENT
HANDLE WITH CARE

6. NOTES.

6.1 Ordering data. - Procurement documents should specify the title, number and date of this specification, and exceptions to this specification, applicable drawings and other documents. Procurement documents should also specify the desired level of packaging and packing.

6.2 Criteria for use of proper level of preservation, packaging and packing shall be as follows:

For Level A. - This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

For Level B. - This level shall be used only when it is definitely known that the item will be held in covered storage in overseas locations for 6 months or less, or in domestic locations for an indefinite period.

For Level C.- This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

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Notice. - When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever: and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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